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AI for Bharat Hackathon

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Team Name : Meson Labs

Team Leader Name : Ashish Sharma

Problem Statement : ReFarm.AI – Smart Plantation Rejuvenation

AI for Plantation Rejuvenation in India

Transforming rubber plantations through intelligent technology

USE CASE: RUBBER PLANTATIONS

Problem Statement

22% Untapped

Significant underutilization of rubber plantation resources across India.

Adaptation Challenges

Farmers struggle with climate adaptation, soil suitability, and crop diversification.

Limited Integration

Lack of connection with mechanized tapping companies reduces efficiency and profitability.

Disconnected Communities

Local communities remain isolated from value chains, limiting socio-economic growth.

ReFarm.AI – Smart Plantation Rejuvenation

An AI-powered solution addressing critical challenges in plantation management

01

AI Agronomy & Diversification Advisor

Climate and soil intelligence models analyze micro-climate, soil health, and rainfall patterns to suggest optimal crop diversification alongside rubber.

02

Mechanized Tapping Integration

AI scheduling optimizes tapping cycles based on weather, tree maturity, and labor availability while enabling digital contracts between farmers and service providers.

03

Community & Cooperative Empowerment

Village AI dashboards track plantation health and market prices, while AI tutors provide training in mechanized tapping and digital literacy.

04

Scalability Beyond Rubber

Modular AI framework adaptable to tea, coffee, cardamom, and other plantation ecosystems with community-led adoption.

Focus on Practical Innovation

Offline-First Design

Local language support ensures rural accessibility without constant internet connectivity.

Community Ownership

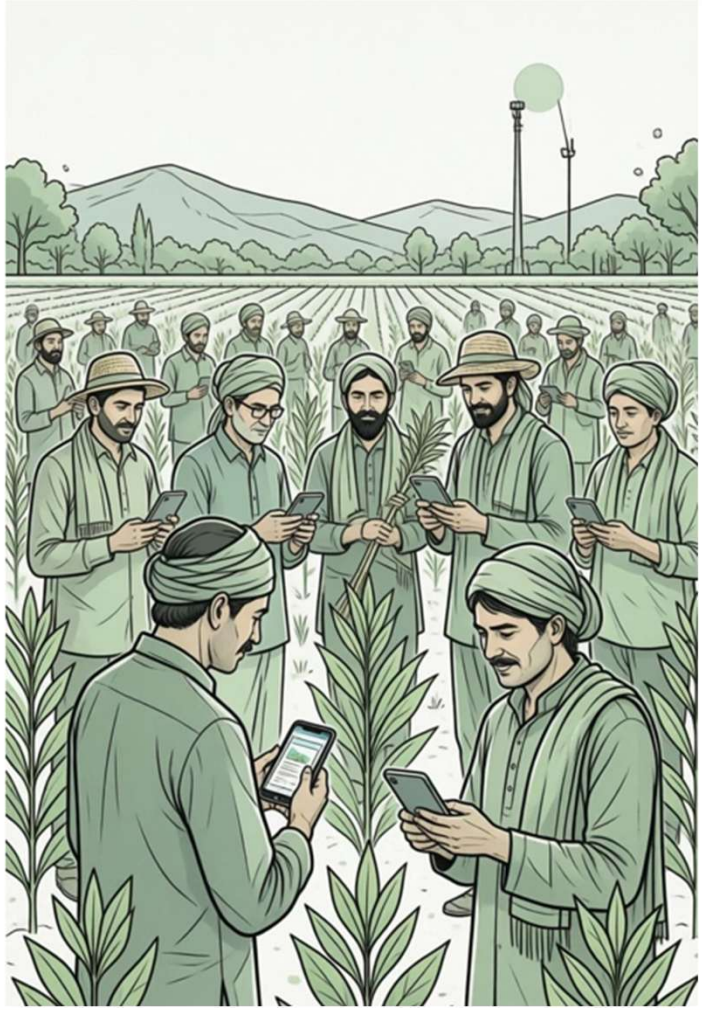
Cooperatives and gram panchayats drive adoption and ensure local control.

IoT + Satellite Integration

Real-time plantation monitoring through connected sensors and satellite data analysis.

Scalable Architecture

Expandable across plantation types and geographies throughout India.



Long-term Societal Value

Creating sustainable transformation across India's plantation sector

Increased Income

Revitalizes untapped plantations, directly boosting farmer earnings.

Stronger Economies

Integrates rural communities into value chains, strengthening local economies.

Climate Resilience

Enhances sustainable land use and adaptation to climate challenges.

Replicable Model

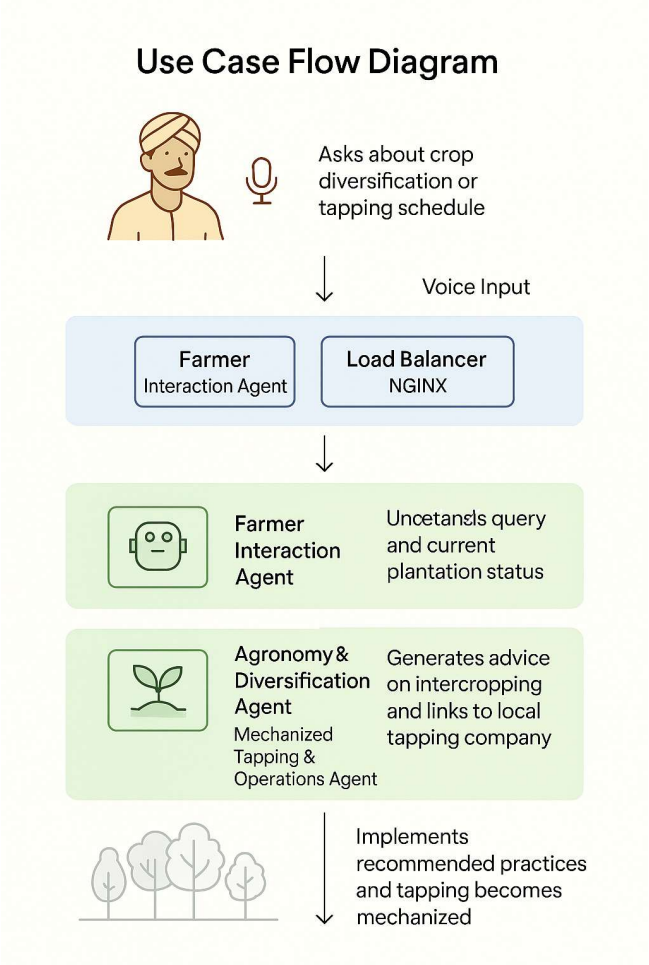
Creates blueprint for transformation across India's entire plantation sector.

Process flow diagram / Use-case diagram

🧭 **Use Case Flow: Voice-Driven Plantation Intelligence**

This diagram illustrates how a farmer's voice query—spoken in Hindi, Tamil, or Malayalam—is processed by ReFarm.AI's multi-agent system to deliver actionable agronomy and tapping recommendations:

- **Voice Input**: The farmer asks about crop diversification or tapping schedules using natural speech.
- **Farmer Interaction Agent**: Interprets the query, retrieves plantation context, and routes it to the right expert agent.
- **Agronomy & Tapping Agents**: Analyze soil, climate, and tree maturity data to generate personalized advice.
- **Community Action**: Recommendations are delivered back in local language, enabling farmers to implement best practices and integrate with mechanized tapping services.



Architecture diagram of the proposed solution:

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#ReFarmAI Technical Stack & Development Guidelines

## Architecture
- **Multi-Agent Framework**: Python (FastAPI), Node.js with standardized
message passing
- **Hybrid Deployment**: Cloud services for complex analytics, edge computing
for real-time responses
- **Voice Processing**: WebRTC, Speech Recognition APIs, Whisper, Festival TTS
- **API-First Design**: RESTful APIs with OpenAPI specifications, WebSocket
for real-time communication
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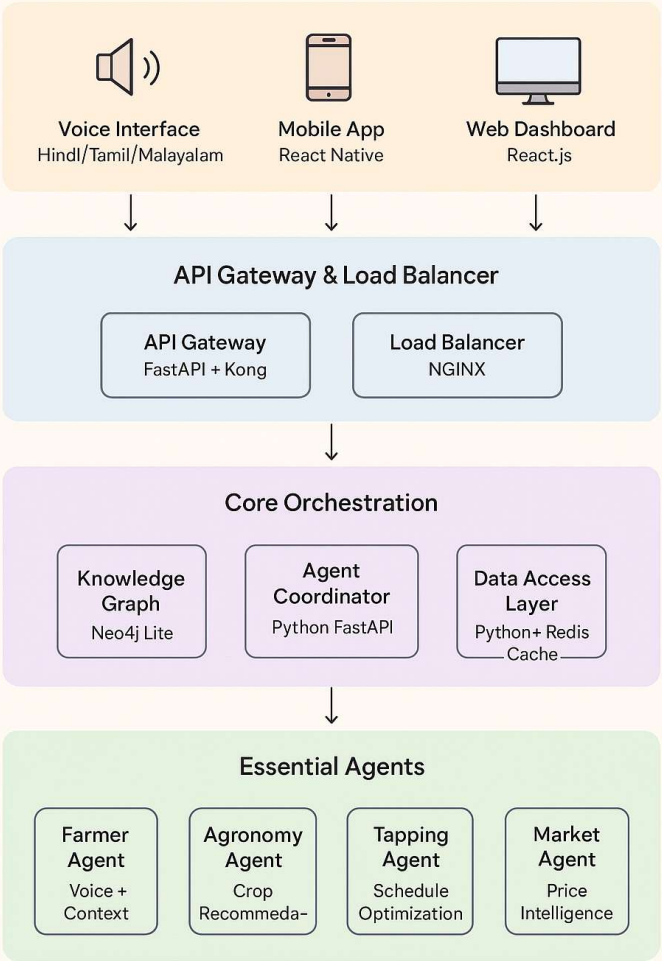

Technologies to be used in the solution:

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## Technology Stack

### Backend & AI/ML
- **Languages**: Python, Node.js, JavaScript
- **Frameworks**: FastAPI, Express.js
- **AI/ML**: TensorFlow, PyTorch, spaCy, Transformers
- **Message Queue**: Apache Kafka, RabbitMQ
- **API Gateway**: Kong, AWS API Gateway

### Data & Storage
- **Time Series**: InfluxDB, TimescaleDB
- **Graph Database**: Neo4j, Amazon Neptune
- **Document Store**: MongoDB, Elasticsearch
- **Relational**: PostgreSQL, MySQL
- **Cache**: Redis, Memcached
- **Data Lake**: For raw data ingestion and batch processing

### Frontend & Mobile
- **Web**: React.js, Vue.js
- **Mobile**: React Native, Flutter
- **Voice Interface**: WebRTC, Speech Recognition APIs
- **Offline Support**: Service Workers, IndexedDB

### Infrastructure & DevOps
- **Containers**: Docker, Kubernetes
- **Cloud**: AWS, Azure, Google Cloud
- **Edge Computing**: AWS IoT Greengrass
- **Monitoring**: Prometheus, Grafana, ELK Stack
- **CI/CD**: Automated pipelines with blue-green deployment
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Thank You